

Bandit CLI Project Analysis

Current Status

This is a well-structured Textual-based CLI application for the OverTheWire Bandit wargame. The project includes:

- SSH terminal interface
- Level information display
- AI mentor system using LiteLLM
- Clean architecture with separation of concerns

Critical Issues Found

1. Missing Requirements File

The README references `requirements.txt` but it's not provided. This is essential for users to install dependencies.

2. AI Model Configuration Issues

- Defaults to `ollama/llama3.2` which requires local Ollama installation
- No fallback mechanism if API fails
- Missing proper error handling for different providers

3. File Path Issues

- Import paths use hardcoded file names instead of proper resource handling
- JSON files might not load correctly in different environments

4. SSH Connection Stability

- No connection keepalive mechanism
- Limited error recovery
- Thread safety concerns with output handling

5. Security Concerns

- SSH host key policy is too permissive (AutoAddPolicy)
- No input validation for SSH parameters
- Passwords stored in memory without clearing

6. User Experience Issues

- No connection status indicators beyond reactive properties
- Limited feedback for long-running operations
- No progress indicators for AI responses

Recommended Improvements

1. Create proper requirements.txt
2. Improve error handling and fallbacks
3. Add input validation
4. Enhance security practices
5. Better user feedback mechanisms
6. Add configuration management with dotenv
7. Implement proper logging

Architecture Assessment

Strengths:

- Good separation of concerns
- Proper use of Textual framework
- Clean class structure
- Reactive programming patterns

Areas for Improvement:

- Error handling needs strengthening
- Configuration management
- Security hardening
- User experience polish